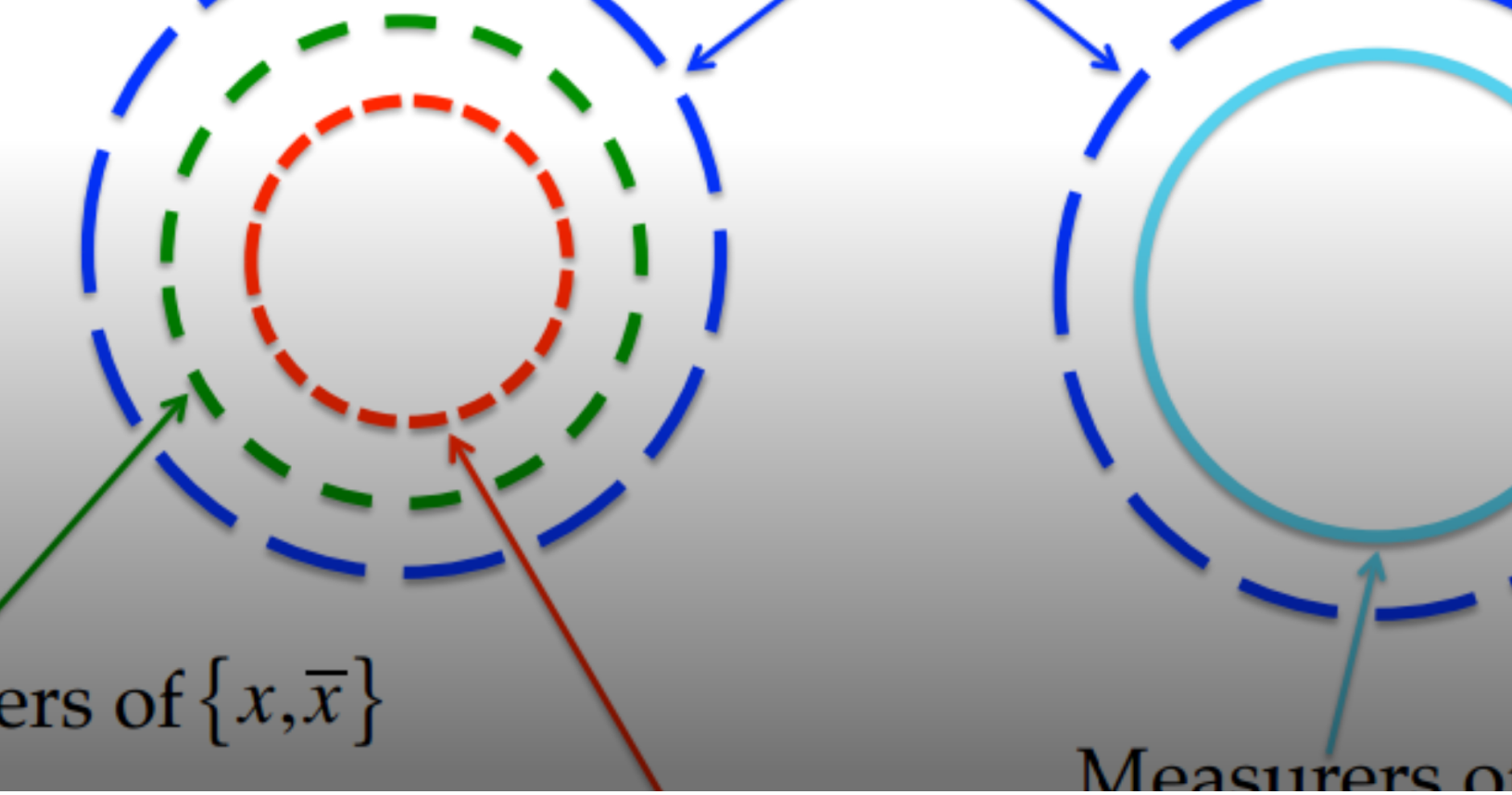



# Deeper Than Quantum Mechanics—David Deutsch's New Theory of Reality

One of the world's leading theorists has a new theory of everything. It's first result: a description of classical and quantum information linking them under the same theoretical umbrella for the first time

One of the unsung heroes of 20th century science is the mathematician and electronics engineer, Claude Shannon, who worked at the famous Bell laboratories during the 1940s, 50s and 60s. Shannon's greatest work is the theory of information which he published in 1948 and has since had a profound influence on our world.

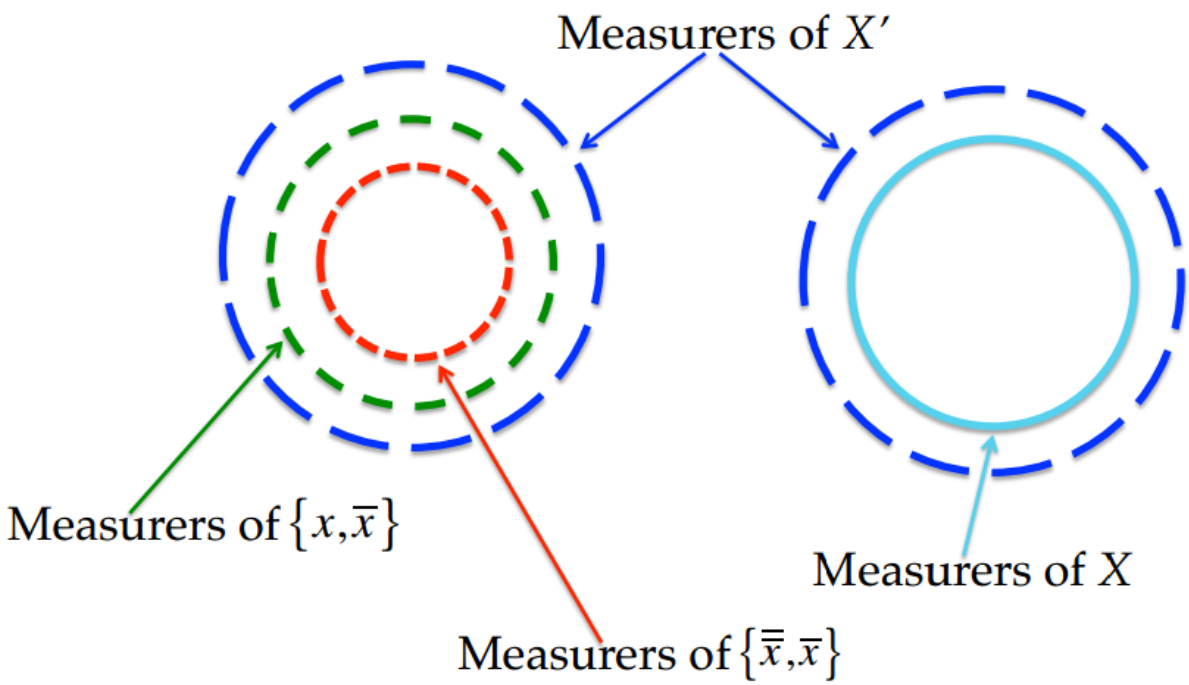
This theory is the basis for all digital communication. So mobile phones, digital television and radio, computers and the Internet all depend on Shannon's theory of information. For that reason, it's possible to argue that Shannon has had a bigger influence on 21st century technology than anybody in history.

But there's a problem his theory of information which has stumped physicists and mathematicians in recent years. This is that it only applies to classical information, the kind of 0s and 1s that make up ordinary digital code.

But physicists have become increasingly interested in quantum information and its potential in cryptography and in quantum computing. Quantum information can be both a 1 and 0 at the same time. This among other exotic properties is what allows quantum computers to be so powerful and quantum cryptography to be perfectly secure.

But Shannon's ideas break down in the quantum regime so various research groups have been searching for an alternative formulation that will give quantum information the same theoretical footing that Shannon gave to its classical cousin.

That goal may now be a step closer thanks to the work of David Deutsch and Chiara Marletto at the University of Oxford in the UK. These guys have come up with a way to link classical and quantum information using a single theory that acts as a foundation for both.




Their new idea is called constructor theory and it is both simpler and deeper than quantum mechanics, or indeed any other laws of physics. In fact, Deutsch claims that constructor theory forms a kind of bedrock of reality from which all the laws of physics emerge.

Constructor theory is a radically different way of thinking about the universe that Deutsch has been developing for some time. He points out that physicists currently ply their trade by explaining the world in terms of initial conditions and laws of motion. This leads to a distinction between what happens and what does not happen.

Constructor theory turns this approach on its head. Deutsch's new fundamental principle is that all laws of physics are expressible entirely in terms of the physical transformations that are possible and those that are impossible.

In other words, the laws of physics do not tell you what is possible and impossible, they are the result of what is possible and impossible. So reasoning about the physical transformations that are possible and impossible leads to the laws of physics.

That's why constructor theory is deeper than anything that has gone before it. In fact, Deutsch does not think about it as a law of physics but as a principle, or set of principles, that the laws of physics must obey.

The analogy that he draws is with conservation laws such as the conservation of energy. This is not a law of physics like quantum mechanics or relativity but a principle that all other laws must follow. As energy is converted from chemical to electrical to kinetic to potential energy and so on, its behaviour is governed by all kinds of different laws of physics. But they must all obey the principle that energy is conserved.

"Thus the conservation law, though not an a priori mathematical truth, provides an explanation of aspects of motion that is deeper than laws of motion," say Deutsch and Marletto.

Constructor theory plays a similar role. "It is a principle, namely a law of physics that expresses and explains constraints on other laws rather than on the behaviour of physical objects directly,"

they say. It is the one law to rule them all, (as...errr...Tolkien might have put it).

Information is similar to energy in this respect. It can be encoded using light, chemistry, electronics, smoke signals and so on, and all these things obey different laws of physics. However, the information itself is somehow separate from all this. It is substrate-independent. But the information itself is preserved, regardless of the laws in play.

The laws or principles that govern the behaviour of information have been unknown, until now. “In this paper we conjecture what these laws are,” say Deutsch and Marletto.

An important point that these guys focus on is that information only exists in physical circumstances—it is never abstract. But that’s in stark contrast to the way that many mathematicians and physicists have approached information in the past.

“Previous attempts to incorporate information at a fundamental level into physics or at least into quantum theory have regarded information as being an a priori mathematical or logical concept,” say Deutsch and Marletto. “Our approach is the opposite.”

Their method is to define a set of nine principles based on constructor theory and apply them to what we know is possible and impossible when it comes to information. These principles express the concepts of computation, measurement and classical information.

Deutsch and Marletto then define a new concept called superinformation in which certain information-related tasks are impossible. They go on to show that the distinctive features of quantum information follow from the impossibility of these tasks. “Quantum information then appears as an instance of superinformation,” they say.

This approach solves a number of problems. In particular, information has always been difficult to define. In conventional information theory, information and distinguishability are each defined in terms of the other, creating a kind of chicken and egg problem.

But in constructor theory, the nature of information is determined by the laws of physics alone. That neatly sidesteps the problem.

It also links quantum information and classical information under the same theoretical roof for the first time. That’s a significant step that could have important implications for the emerging technology of quantum computation, cryptography and communication.

It’s important to point out that constructor theory is not a way of deriving the laws of physics. Deutsch and Marletto are not attempting, for example, to derive quantum mechanics from some deeper theory.

Instead, the principles function very much like the conservation of energy. The conservation of energy is not a mathematical truth

but Deutsch and Marletto say it is deeper than the laws of physics that obey it. By this, they mean that any as-yet-undiscovered laws must also be expected to obey this conservation rule.

The principles from constructor theory work in the same way. The known laws of physics obey these principles and any unknown, yet-to-be-discovered laws must too.

The big unanswered question about constructor theory is how useful it will turn out to be. Deutsch is fascinated by the fundamental properties of reality and for him a deeper explanation is reason enough to explore further.

Others will demand more—testable hypotheses, for example, that can determine whether constructor theory is true or not. These kinds of predictions will surely emerge as more physicists discover Deutsch's new way of thinking.

And physicists will surely be tempted to explore this idea further not least because Deutsch is widely acknowledged as one of the leading thinkers on the foundations of physics and one of the most creative and unconventional too.

He pioneered the ideas behind quantum computing in the 1980s. At that time, the notion of using quantum mechanics for calculations was a backwater of physics. Today, it is one of the driving forces not just of new physics but of new technology as well.

Later, he became one of the main exponents of the many-worlds interpretation of quantum mechanics and the multiverse. That too has morphed from a minority view to a mainstream idea in cosmology.

In other words, Deutsch has an impressive track record.

Only a fool would bet against the possibility that constructor theory could also become a mainstream idea in physics that will have profound consequences for our future understanding of the universe. Shannon, who died in 2001, would surely be impressed.

Ref: arxiv.org/abs/1405.5563 : Constructor Theory of Information

WRITTEN BY

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Published May 28, 2014

PUBLISHED IN

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